

LONS-LE-SAUNIER, France

WMO: 073900

Lat: 46.6928N

Lon: 5.5177E

Elev: 915

StdP: 14.22

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.2	22.4	13.5	11.3	22.8	17.4	13.7	24.4	25.6	47.2	22.5	45.4	3.1	100	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	88.7	68.7	85.1	68.2	82.1	67.4	71.3	83.3	69.7	81.9	68.3	79.6	6.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	103.0	77.9	65.5	97.3	75.5	64.1	92.6	73.7	35.6	83.8	34.2	82.3	33.1	80.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	17.9	14.2	15.2	92.7	4.6	5.0	11.9	96.3	9.2	99.3	6.6	102.1	3.2	105.8	
			DB												
			WB	14.4	74.0	4.4	1.6	11.2	75.2	8.6	76.1	6.2	77.0	3.0	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	36.8	40.0	45.9	52.2	60.1	66.3	68.5	67.3	62.2	55.2	44.2	37.8
	DBStd	13.45	8.26	7.45	7.60	6.55	6.38	7.24	6.35	6.27	6.41	6.87	7.71	8.30
	HDD50	1543	413	285	170	51	4	0	0	0	1	30	204	385
	HDD65	4780	874	699	594	384	176	73	35	41	126	309	625	844
	CDD50	2677	4	6	41	118	317	490	573	537	367	190	29	5
	CDD65	442	0	0	0	1	25	113	143	113	42	5	0	0
	CDH74	3693	0	0	0	19	207	994	1244	978	239	12	0	0
	CDH80	1139	0	0	0	0	32	290	420	367	30	0	0	0
Wind	WSAvg	5.9	5.8	6.5	7.0	6.9	6.1	5.5	5.0	4.6	5.4	5.8	5.5	6.2
Precipitation	PrecAvg	54.3	4.4	4.0	3.8	4.4	4.8	4.3	4.2	4.2	4.3	5.5	5.9	4.9
	PrecMax	69.4	9.1	9.4	13.3	9.6	8.1	8.3	9.8	9.6	9.9	10.0	16.7	8.9
	PrecMin	42.1	1.8	0.9	0.7	0.8	1.7	1.1	1.8	0.6	1.2	1.7	1.4	0.2
	PrecStd	7.4	2.0	2.1	2.6	2.4	1.9	1.9	2.2	2.4	2.3	3.3	3.3	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	59.9	69.1	77.0	84.0	91.3	93.3	99.8	83.6	75.7	65.4	59.7
		MCWB	50.1	49.4	55.5	60.5	66.8	68.3	67.7	67.6	66.0	63.9	56.4	52.4
	2%	DB	53.9	56.6	64.2	71.5	79.2	86.1	88.1	88.9	80.2	71.0	61.7	55.2
		MCWB	49.3	48.2	52.2	57.9	64.2	69.1	69.4	67.4	66.6	61.4	54.0	49.5
	5%	DB	52.0	53.3	60.3	68.0	75.7	83.2	84.8	83.9	76.9	68.4	57.9	52.2
		MCWB	47.6	46.8	50.4	55.6	62.5	67.9	68.4	67.7	64.7	60.0	52.4	48.3
	10%	DB	49.0	50.8	57.3	63.9	72.4	80.1	81.2	79.3	73.1	65.5	55.0	49.6
		MCWB	45.5	45.6	49.7	53.9	60.8	66.5	67.4	66.3	63.1	58.9	50.9	46.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.0	56.9	61.8	68.9	73.0	73.4	71.7	70.7	65.3	57.9	53.0
		MCDB	54.7	56.4	68.2	73.7	80.0	85.0	85.6	83.5	80.1	72.3	62.9	57.1
	2%	WB	49.8	49.2	53.9	58.8	65.9	71.2	71.3	69.9	68.4	63.0	54.9	50.4
		MCDB	53.6	54.5	60.6	69.3	77.2	82.8	84.1	83.7	77.1	68.6	59.5	54.0
	5%	WB	47.8	47.6	52.2	56.7	63.8	69.2	69.5	68.4	66.1	61.1	53.0	48.5
		MCDB	51.3	52.5	58.2	66.4	73.4	80.5	81.7	81.2	74.0	67.2	57.6	52.1
	10%	WB	45.7	46.1	50.4	54.5	61.7	67.3	67.8	67.1	63.9	59.1	51.0	46.5
		MCDB	48.8	50.5	56.6	62.7	70.2	77.6	79.1	78.1	71.0	64.7	54.8	49.1
Mean Daily Temperature Range	5% DB	MDBR	9.9	11.8	14.8	16.5	17.0	18.5	18.1	17.5	16.7	13.4	10.7	8.8
		MCDBR	11.4	15.1	19.9	23.1	22.1	22.3	22.9	24.2	20.2	16.4	13.8	11.0
		MCWBR	8.0	9.6	11.4	12.7	11.5	10.3	9.8	10.4	9.8	9.3	8.7	7.7
	5% WB	MCDBR	10.6	12.8	16.3	21.4	19.9	21.1	21.5	21.2	18.1	15.3	12.7	10.7
		MCWBR	7.7	8.6	9.7	12.1	10.9	10.2	10.0	10.3	9.9	9.5	8.8	8.0
Clear-Sky Solar Irradiance	taub	0.308	0.329	0.361	0.396	0.402	0.407	0.404	0.398	0.377	0.362	0.328	0.306	
	taud	2.447	2.389	2.335	2.250	2.280	2.303	2.308	2.339	2.378	2.411	2.454	2.457	
	Ebn at Noon	250	266	273	272	274	272	272	269	264	250	239	236	
	Edh at Noon	22	28	34	41	41	40	39	37	32	27	22	19	
All-Sky Solar Radiation	RadAvg	356	619	1026	1415	1653	1945	1864	1614	1245	769	410	291	
	RadStd	44	98	137	197	172	176	161	133	98	64	39	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page